Hashir Aziz

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Education

National University of Sciences and Technology (NUST) — Islamabad, Pakistan
Bachelor of Science in Computer Science | CGPA: 3.7 — *2023 – 2027*

FG Sir Syed College (F.Sc Pre-Engineering) — Rawalpindi, Pakistan
Higher Secondary School Certificate | 1001/1100 Marks — *2021 – 2023*

Projects

C++ Search Engine | C++, libcurl, libxml2, pybind11, nlohmann/json

- Designed and implemented a multithreaded web crawler in C++ using a BFS approach, with proper handling of `robots.txt` and crawl depth limits.
- Extracted keywords by prioritizing content inside `<h1>`, `<h2>`, and `<p>` tags; removed stopwords and applied lemmatization to normalize terms.
- Mapped keywords to URLs using hash maps for easy indexing and retrieval.
- Built an indexing engine that ranked pages using a combination of PageRank and TF-IDF, enabling contextual and relevant search results.
- Indexed up to 2000 pages with query time of a few seconds for lengthy queries.

Music Streaming Platform with Personalized Recommendation Engine | MERN Stack

- Built a full-stack music app using the MERN stack with core features like login, song playback, playlist creation, and user analytics.
- Designed a modular backend system that tracked user listening behavior and generated personalized song recommendations.
- Optimized performance using smart thresholds to trigger recommendations at the right time, reducing overhead and improving responsiveness.
- Implemented a dynamic user profile page showing favorite genres/artists and listening patterns via bar charts.
- Wrote individual tests for all different sections of the recommendation system.

Checkers Game with AI | Python, Pygame, Alpha-Beta Pruning

- Created a desktop-based checkers game supporting two-player mode and AI opponent.
- Implemented an AI using Minimax with Alpha-Beta pruning to improve decision-making speed and strategy.
- Included adjustable difficulty levels and a user-friendly interface for smooth gameplay.

Digital Logic Design Simulator | Java, JavaFX

- Built a visual simulation engine in Java that allows users to build and test digital circuits using standard logic gates.
- Supported real-time simulation with drag-and-drop interaction and customizable gate connections.
- Helped deepen my understanding of Object-Oriented Programming by applying concepts like encapsulation, inheritance, and polymorphism in a real-world simulation tool.

Technical Skills

Languages: Python, JavaScript, C/C++, Java, SQL, HTML/CSS, MongoDB Query Language
Frameworks: React, Node.js, Express, Django, Three.js
Tools: Git, GitHub, MongoDB, IntelliJ, VS Code, WebStorm
Concepts: OOP, REST APIs, TF-IDF, PageRank, DSA, Multithreading